CAMPANA POINTS ON QUADRICS WITH HYPERPLANE BOUNDARY

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Abstract. Let $Q \subset \mathbf{P}_k^{d+1}$ be a smooth quadric of dimension $d \geq 2$ over a number field, and let $H_i = Q \cap \Pi_i$ be hyperplane sections, for $1 \leq i \leq d$. For every integer $m \geq 2$, we prove Campana strong approximation for the pair $\big(Q, (1-1/m)(H_1 + \cdots + H_d)_{\mathrm{red}}\big)$.

1. Introduction

Quadrics form a classical family in number theory. By the Hasse–Minkowski theorem, they satisfy the Hasse principle and weak approximation. For affine quadrics of dimension at least 3, strong approximation holds under a non-compactness condition; in dimension 2, it holds with the Brauer–Manin obstruction [CTX09, Theorem 8.3]. The Hasse principle and weak approximation for intersections of two quadrics have also been extensively investigated; see, for instance, [CTSSD87a, CTSSD87b, SS91, Wit07, HB18, CT24].

In the Campana setting, Nakahara and Streeter established approximation results for projective spaces, including for the projective line with a boundary of degree two [NS24]; The log Fano case on $\mathbf{P}^1$ is treated in [Wei26b, Theorem 1.1]. Mitankin, Nakahara and Streeter proved Campana strong approximation for a smooth quadric whose boundary is a single hyperplane section [MNS24, Theorem 1.4]. In this paper, we treat the case where the boundary is the reduced union of hyperplane sections.

For $d \geq 2$, let $Q \subset \mathbf{P}_k^{d+1}$ be a smooth quadric, let $\Pi_1, \ldots, \Pi_d$ be hyperplanes defined over k, and put

$$H_i = Q \cap \Pi_i, \qquad D = H_1 + \cdots + H_d, \qquad \Delta_m = \left(1 - \frac{1}{m}\right) D_{\mathrm{red}}. \tag{1.1}$$

Each distinct prime component of D occurs in D_{red} with multiplicity one and the hyperplanes need not be distinct. For $d \geq 2$, every hyperplane section is geometrically reduced, since its quadratic form has rank at least d. On a surface, distinct sections may share a line. For $d \geq 3$ it is geometrically integral. This pair is of Fano type since

$$-(K_Q + \Delta_m) \sim \frac{d}{m} \mathcal{O}_Q(1) + \left(1 - \frac{1}{m}\right)(D - D_{\mathrm{red}})$$

is ample.

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Theorem 1.1. *Let $Q \subset \mathbf{P}_k^{d+1}$ be a smooth quadric of dimension $d \geq 2$. let $\Pi_1, \dots, \Pi_d$ be hyperplanes defined over k, and let D, Δ_m be defined as in (1.1). Then, for every integer $m \geq 2$, the pair (Q, Δ_m) satisfies Campana strong approximation; that is*

$$\overline{(Q \setminus D)(k)} = (Q, \Delta_m)(\mathbf{A}_k). \tag{1.2}$$

Here the closure is taken in the Campana topology of Definition 2.1.

As an immediate consequence of Theorem 1.1, we obtain the following corollary concerning S-Campana points.

Corollary 1.2. *Under the hypotheses of Theorem 1.1, fix a finite $S \supseteq \Omega_k^\infty$ and a proper flat integral $\mathcal{O}_S$-model $\mathscr{Q}$ of (Q, Δ_m). The strict S-Campana points of this model are dense in*

$$\mathcal{A}_S(\mathscr{Q}, \Delta_m) = \prod_{v \in S} Q(k_v) \times \prod_{v \notin S} \mathcal{C}_v(\mathscr{Q}, \Delta_m) \tag{1.3}$$

with its product topology, where $\mathcal{C}_v$ is defined using this fixed model.

By monotonicity of Campana strong approximation [Wei26a, Lemma 2.5(1)], we immediately obtain the following result for unequal weights.

Corollary 1.3. *Under the hypotheses of Theorem 1.1, write $D_{\mathrm{red}} = \sum_{\alpha \in I} D_\alpha$ for its distinct prime components. Let $n_\alpha \geq 2$ be arbitrary integers and put*

$$\Delta_{\boldsymbol{n}} = \sum_{\alpha \in I} \left(1 - \frac{1}{n_\alpha}\right) D_\alpha. \tag{1.4}$$

Then $(Q, \Delta_{\boldsymbol{n}})$ satisfies Campana strong approximation.

The common intersection

$$E = Q \cap \Pi_1 \cap \cdots \cap \Pi_d$$

is understood scheme-theoretically. First suppose that $E(k) \neq \varnothing$, maybe E is not finite. Let $L = \Pi_1 \cap \cdots \cap \Pi_d$. Choose a line $\ell \subset L$ through a rational point of E. Let Π be a general plane containing ℓ. If $\ell \subset Q$, then $\Pi \cap Q = \ell \cup C$, where C is the residual line, whose reduced boundary is the single point $x = C \cap \ell$. Otherwise, $C = \Pi \cap Q$ is a smooth conic with reduced boundary $(C \cap \ell)_{\mathrm{red}}$ of degree at most two. Proposition 3.2 treats all these cases.

We may therefore assume $E(k) = \varnothing$. When E is finite, the defining linear forms Π_i are linearly independent. Then $\Lambda = \Pi_1 \cap \cdots \cap \Pi_d \simeq \mathbf{P}_k^1$ is not contained in Q, D is reduced, and E has length two. It is a closed point of degree two. For surfaces, we use the conic pencil through $H_1 \cap H_2$. By a fibration result in [HWW22], we find a smooth fiber with local points in the prescribed Campana open neighborhood. Campana strong approximation on $\mathbf{P}^1$ then give the desired rational point. For $d > 2$, the common plane conic is automatically reduced. We may choose a suitably quadric surface through $Q \cap \Pi_2 \cap \cdots \cap \Pi_d$, which then reduces the approximation problem to the surface case.

It remains to treat positive-dimensional E with $E(k) = \varnothing$. Choose a plane $L' \subset L$ with smooth conic $B = Q \cap L'$. For general R, the section $\Sigma_R = Q \cap \langle L', R \rangle$ is a smooth

quadric surface, and every H_i restricts to B. The one-boundary surface case in Corollary 4.4 completes this reduction.

The paper is organised as follows. After the preliminaries in Section 2, Section 3 treats all E with $E(k) \neq \varnothing$. The remanining surface case, the hardest part of the paper,is proved in Section 4. Section 5 carries out the remaining reductions to surfaces, for finite and positive-dimensional E.

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2. Campana approximation and preliminary results

Let Ω_k be the set of places of k and Ω_k^∞ the archimedean places. For a finite place v, we write $\mathcal{O}_v$, $\mathfrak{m}_v$ and $\mathbf{F}_v$ for its valuation ring, maximal ideal and residue field. We normalize ord_v to have image $\mathbf{Z}$. Throughout this paper, we always assume the finite set of places $S \supseteq \Omega_k^\infty$. Let $\mathcal{O}_S$ be the ring of S-integers.

2.1. **Campana topology.** Let X be a smooth geometrically integral k-variety, not necessarily proper. Let $D = \sum_i D_i$ be a reduced divisor with distinct prime components defined over k, and put

$$\Delta = \sum_i \left(1 - \frac{1}{m_i}\right) D_i, \qquad m_i \in \mathbf{Z}_{\geq 2}.$$

All weights on X are finite. For equal weights $m_i = m$, this gives $\Delta = (1 - 1/m)D$. Fix a finite set S and a flat separated integral $\mathcal{O}_S$-model $\mathscr{X}$ of X. Let $\mathscr{D}_i$ be the reduced closure of D_i.

For $v \notin S$, write $D_i \times_k k_v = \bigcup_\lambda D_{i,\lambda}$ for the local irreducible components, and let $\mathscr{D}_{i,\lambda}$ be their reduced closures in $\mathscr{X}_{\mathcal{O}_v}$. For $x \in \mathscr{X}(\mathcal{O}_v) \subset X(k_v)$, let $\widetilde{x} : \operatorname{Spec} \mathcal{O}_v \to \mathscr{X}_{\mathcal{O}_v}$ be its section.

For any closed subscheme $\mathscr{Z} \subset \mathscr{X}_{\mathcal{O}_v}$, let $\mathcal{I}_{\mathscr{Z}} \subset \mathcal{O}_{\mathscr{X}_{\mathcal{O}_v}}$ be its ideal sheaf. Define its pulled-back ideal along $\widetilde{x}$ by

$$J_{\mathscr{Z},x} = \widetilde{x}^{-1}\mathcal{I}_{\mathscr{Z}} \cdot \mathcal{O}_v \subseteq \mathcal{O}_v.$$

If $f_1, \ldots, f_r$ generate $\mathcal{I}_{\mathscr{Z}}$ near the reduction of x, then $J_{\mathscr{Z},x} = (f_1(x), \ldots, f_r(x))$. In particular, write $J_{i,\lambda,x} := J_{\mathscr{D}_{i,\lambda},x}$. Since $\mathcal{O}_v$ is a discrete valuation ring, a nonzero $J_{i,\lambda,P}$ equals $\mathfrak{m}_v^n$ for a unique integer $n \geq 0$. We define the *intersection multiplicity*

$$n_v(\mathscr{D}_{i,\lambda}, x) = \begin{cases} \mathrm{length}_{\mathcal{O}_v}(\mathcal{O}_v / J_{i,\lambda,x}), & J_{i,\lambda,x} \neq 0, \\ \infty, & J_{i,\lambda,x} = 0. \end{cases} \tag{2.1}$$

Define

$$\mathcal{C}_v(\mathscr{X}, \Delta) = \{x \in \mathscr{X}(\mathcal{O}_v) : n_v(\mathscr{D}_{i,\lambda}, x) \in \{0\} \cup \mathbf{Z}_{\geq m_i} \cup \{\infty\} \text{ for every } i, \lambda\}, \qquad v \notin S,$$
$$\mathcal{C}_v(\mathscr{X}, \Delta) = X(k_v), \qquad v \in S.$$

We use the local-component convention of [NS24, Definitions 2.2–2.4]. A local point $x \in \mathcal{C}_v(\mathscr{X}, \Delta)$ is called a *strict local Campana point* if $x \in (X \setminus D)(k_v)$. A rational point

$x \in (X \setminus D)(k)$ is called a *strict S-Campana point* for the fixed model if it extends to an $\mathcal{O}_v$-point of $\mathscr{X}$ for every $v \notin S$ and satisfies

$$n_v(\mathscr{D}_{i,\lambda}, x) = 0 \quad \text{or} \quad n_v(\mathscr{D}_{i,\lambda}, x) \geq m_i \qquad \text{for every } v \notin S,\ i \text{ and } \lambda.$$

For $\mathcal{O}_S$-model and a finite set $T \supset S$, write

$$\mathcal{A}_T(\mathscr{X}, \Delta) = \prod_{v \in T} X(k_v) \times \prod_{v \notin T} \mathcal{C}_v(\mathscr{X}, \Delta)$$

with the product topology.

Definition 2.1 (Campana topology)**.** The *Campana adelic space* is

$$(X, \Delta)(\mathbf{A}_k) = \prod_v{}' (X(k_v), \mathcal{C}_v(\mathscr{X}, \Delta)) = \bigcup_{T \supseteq S \text{ finite}} \mathcal{A}_T(\mathscr{X}, \Delta). \tag{2.2}$$

Its points satisfy the local Campana conditions at almost every place. We equip it with the restricted-product topology, which we call the *Campana topology*. It is independent of the choice of the model since any two models agree outside finitely many places. A basis consists of

$$\prod_{v \in T} W_v \times \prod_{v \notin T} \mathcal{C}_v(\mathscr{X}, \Delta), \qquad T \supseteq S \text{ finite}, \tag{2.3}$$

where $W_v \subset X(k_v)$ is open in the usual local adelic topology. Following [Wei26a, Definition 2.1], the pair (X, Δ) satisfies *Campana strong approximation* if $(X \setminus D)(k)$ is dense in $(X, \Delta)(\mathbf{A}_k)$.

We recall [Wei26a, Lemma 2.3 and Proposition 2.4(1)].

Lemma 2.2. (1) *For finite weights, $\mathcal{C}_v(\mathscr{X}, \Delta)$ is open and closed in $X(k_v)$.*
(2) *Campana strong approximation for (X, Δ) is equivalent to Campana weak approximation for every finite S and every flat separated integral model; see* [NS24, Definition 2.7(ii)].

2.2. **Orbifold morphisms and continuity.**

Definition 2.3 (Orbifold morphism)**.** Let (Y, Δ_Y) and (X, Δ_X) be smooth geometrically integral k-varieties with finite-weight Campana boundaries. Each geometric boundary component inherits its weight. Let $m_Y(F) = 1$ for a geometric prime divisor F outside the boundary of $Y_{\overline{k}}$. Following [Cam11, Definition 2.3], a k-morphism $f : Y \to X$ is a *morphism of Campana orbifolds* if $f(Y) \not\subseteq \operatorname{Supp} \Delta_X$ and, for every geometric boundary component D of $X_{\overline{k}}$,

$$f_{\overline{k}}^* D = \sum_F e_{F,D} F \quad (e_{F,D} > 0) \quad \Longrightarrow \quad e_{F,D}\, m_Y(F) \geq m_X(D). \tag{2.4}$$

Here $f_{\overline{k}}^* D$ is the effective Cartier pullback of D. The sum runs over the prime components of its support, and $e_{F,D}$ is the order of F. For an inclusion $j : Y \hookrightarrow X$, this is the scheme-theoretic divisor restriction $D|_Y$, with its multiplicities.

Proposition 2.4 (Continuity)**.** *An orbifold morphism $f : (Y, \Delta_Y) \to (X, \Delta_X)$ induces a continuous map*

$$f_{\mathbf{A}} : (Y, \Delta_Y)(\mathbf{A}_k) \longrightarrow (X, \Delta_X)(\mathbf{A}_k).$$

More precisely, for any fixed models $\mathscr{Y}$ of Y and $\mathscr{X}$ of X, there is a finite set $T \supseteq \Omega_k^\infty$, such that

$$f(\mathcal{C}_v(\mathscr{Y}, \Delta_Y)) \subseteq \mathcal{C}_v(\mathscr{X}, \Delta_X) \qquad (v \notin T). \tag{2.5}$$

In particular, the inverse image of a Campana-open set is open.

Proof. Choose a finite Galois extension L/k over which all geometric components of the boundaries and of the divisors $f_{\overline{k}}^* D$ are defined. Choose a finite T containing the ramified places of L/k, so that f extends to the models and the pullback identities hold for the flat Cartier closures over the integral closure of $\mathcal{O}_T$ in L. For $v \notin T$ and $w \mid v$, the k_v-irreducible components correspond to $\mathrm{Gal}(L_w/k_v)$-orbits of the geometric components. For a local irreducible component D_λ of the boundary on X, write $f^* D_\lambda = \sum_\alpha e_{\alpha,\lambda} F_\alpha$ ($e_{\alpha,\lambda} > 0$). The geometric inequalities imply $e_{\alpha,\lambda} m_Y(F_\alpha) \geq m_X(D_\lambda)$. For $x \in \mathcal{C}_v(\mathscr{Y}, \Delta_Y)$ and $v \notin T$,

$$n_v(\mathscr{D}_\lambda, f(x)) = \sum_\alpha e_{\alpha,\lambda}\, n_v(\mathscr{F}_\alpha, x). \tag{2.6}$$

Every positive $n_v(\mathscr{F}_\alpha, x)$ is at least $m_Y(F_\alpha)$. Hence every positive $n_v(\mathscr{D}_\lambda, f(P))$ is at least $m_X(D_\lambda)$. This proves the inclusion (2.5). The inclusion shows that $f_{\mathbf{A}}$ is well-defined. To prove continuity, let $\mathcal{N} = \prod_v N_v$ be a basic neighborhood of $f_{\mathbf{A}}((y_v))$ in $(X, \Delta_X)(\mathbf{A}_k)$. Choose a finite B containing T, the prescribed places of $\mathcal{N}$, and all places where $y_v \notin \mathcal{C}_v(\mathscr{Y}, \Delta_Y)$. At $v \in B$, local continuity gives an open neighborhood U_v of y_v with $f(U_v) \subseteq N_v$. Then

$$\prod_{v \in B} U_v \times \prod_{v \notin B} \mathcal{C}_v(\mathscr{Y}, \Delta_Y) \subseteq f_{\mathbf{A}}^{-1}(\mathcal{N})$$

is a basic neighborhood of (y_v) in $(Y, \Delta_Y)(\mathbf{A}_k)$. This proves continuity. □

2.3. **Approximation results.** We use the following result on curves.

Proposition 2.5 (Nakahara–Streeter)**.** *Let $C \simeq \mathbf{P}^1_k$, and let $E \subset C$ be a reduced divisor of degree at most two. Assign to each closed point $P \in \mathrm{Supp}(E)$ an arbitrary integer $n_P \geq 2$ and put*

$$\Delta_E = \sum_{P \in \mathrm{Supp}(E)} \left(1 - \frac{1}{n_P}\right) P.$$

Then (C, Δ_E) satisfies Campana strong approximation.

Proof. If $E = 0$, use ordinary weak approximation on $\mathbf{P}^1$. If every point of E is rational, apply [NS24, Proposition 3.5] with $n = 1$. Otherwise E is a single closed point of degree two, and [NS24, Proposition 3.15] applies. Lemma 2.2 gives Campana strong approximation. For a more general statement, see [Wei26b, Theorem 1.1]. □

We also use ordinary strong approximation on affine cones in the following case.

Theorem 2.6 (Wei–Xu–Zhu)**.** *Let $V \subset \mathbf{P}^n_k$, $n \geq 7$, be a pure geometrically integral intersection of two quadrics, not a cone, and suppose that V has finitely many singular points. Let W be its affine cone and let $Z \subset W$ be closed of codimension at least two. Then $W^{\mathrm{sm}} \setminus Z$ satisfies strong approximation off any one place of k.*

This is [WXZ25, Theorem 1.6(iii)]. In particular, no Brauer–Manin condition is required in this dimension.

We use the following version of the formal lemma for Campana points [MNS24, Theorem 3.17], a variant of Harari's formal lemma [Har94, Corollary 2.6.1]. We follow the local-component convention of Section 2.1 and the Campana topology of Definition 2.1. In the compactification version, coefficient one is assigned to the divisors at infinity (in $Y^c \setminus Y$), and finite auxiliary weights to additional omitted divisors (in $Y \setminus V$ but not in D).

Lemma 2.7. *Let Y be smooth and geometrically integral, and let Δ have finite weights. Let $V \subset Y \setminus \operatorname{Supp}\Delta$ be a dense open subset, and let $\mathcal{B} \subset \operatorname{Br} V$ be a finite subgroup. Suppose $(p_v) \in (Y,\Delta)(\mathbf{A}_k)$ is orthogonal to $\mathcal{B} \cap \operatorname{Br} Y$. For every neighborhood $\mathcal{N}$ of (p_v) in the Campana topology, there exists*

$$(p'_v) \in (Y,\Delta)(\mathbf{A}_k) \cap V(\mathbf{A}_k)^{\mathcal{B}} \quad \text{with} \quad (p'_v) \in \mathcal{N}.$$

Thus (p'_v) can be chosen arbitrarily close to (p_v) in the Campana topology.

3. The case $E(k) \neq \emptyset$

Let $Q \subset \mathbf{P}(V)$ and $H_1, \dots, H_d$ be as in Theorem 1.1, and let q be the nondegenerate quadratic form defining Q. Write

$$L = \bigcap_{i=1}^{d} \Pi_i = \mathbf{P}(W), \qquad E = Q \cap L.$$

We do not assume that E is finite.

Let $\ell \subset L$ be a k-line. If $\ell \subset Q$, then for general $R \in Q \setminus D$ the plane $\langle \ell, R\rangle$ is not contained in Q, and

$$Q \cap \langle \ell, R\rangle = \ell \cup C_R,$$

where C_R is a distinct k-line through R. Every Π_i cuts this plane along ℓ, so $H_i|_{C_R} = P_R$, where $P_R = \ell \cap C_R \in E(k)$.

If $\ell \not\subset Q$, put $Z = Q \cap \ell$. For general $R \in Q \setminus D$, the plane $\langle \ell, R\rangle$ cuts out a smooth conic C_R, and $H_i|_{C_R} = Z$ for every i. Indeed, the restriction of q to the two-dimensional space defining ℓ has rank two or one. In rank two, adjoining a vector of nonzero norm in its orthogonal complement gives a nondegenerate three-dimensional restriction. The resulting smooth conic contains a geometric point $R \notin \ell$. Thus the smoothness locus in $Q \setminus \ell$ is nonempty; it meets $Q \setminus D$ by irreducibility of Q. In rank one, $Z = 2P$ for some $P \in Q(k)$, and the conic is smooth whenever $R \notin T_P Q$.

Lemma 3.1 (Reduction to a rational curve)**.** *Let $C \subset Q$ be a smooth plane conic or a k-line, not contained in D, and suppose $F = (D|_C)_{\mathrm{red}}$ has degree at most two. Put $\delta_C = (1-1/m)F$*

and let $j : C \hookrightarrow Q$. If a Campana-open set $\mathcal{N} \subset (Q, \Delta_m)(\mathbf{A}_k)$ has nonempty inverse image

$$\mathcal{G} = j_{\mathbf{A}}^{-1}(\mathcal{N}) \subset (C, \delta_C)(\mathbf{A}_k),$$

then $\mathcal{N}$ contains a point of $(C \setminus F)(k)$.

Proof. Each geometric boundary pullback is supported on F, so j is an orbifold morphism and $\mathcal{G}$ is open by Proposition 2.4. Its nonemptiness gives local points on C everywhere; the Hasse principle yields $C \simeq \mathbf{P}^1_k$ when C is a conic. Proposition 2.5 gives a strict rational point in $\mathcal{G}$, whose image lies in $\mathcal{N}$. □

Proposition 3.2. *Under the hypotheses of Theorem 1.1, assume that $E(k) \neq \varnothing$. Then (Q, Δ_m) satisfies Campana strong approximation for every $m \geq 2$.*

Proof. Choose $A \in E(k)$ and a k-line $\ell \subset L$ through A. For general $R \in (Q \setminus D)(k)$, let $C = C_R$ be the residual line if $\ell \subset Q$, and the smooth conic $Q \cap \langle \ell, R \rangle$ otherwise. Put $F = (D|_C)_{\mathrm{red}}$. In the line case $F = \ell \cap C$ is one rational point; in the conic case $F = (Q \cap \ell)_{\mathrm{red}}$ has degree at most two, and $A \in F(k)$.

Let $\mathcal{N} \subset (Q, \Delta_m)(\mathbf{A}_k)$ be nonempty and open. Shrink $\mathcal{N}$ to the basic form (2.3), with local factors W_v at a finite set T, large enough that any point in $F(k)$ satisfies the local conditions outside T. Using weak approximation on Q, we may choose $R \in W_v \cap Q(k)$ for all $v \in T$ and some $P \in F(k)$ also satisfies these conditions outside T. If $\ell \not\subset Q$, take $P = A$. If $\ell \subset Q$ and $d \geq 3$, $P = \ell \cap C$ lies on every boundary component, since all H_i are geometrically integral. The tuple equal to R at T and to P elsewhere belongs to $(C, (1 - 1/m)F)(\mathbf{A}_k)$ and maps into $\mathcal{N}$. The proof then follows from Lemma 3.1.

For the remaining case: $d = 2$ and $\ell \subset Q$, write $H_i = \ell + C_{s_i}, i = 1, 2$. Let $f : Q \to \mathbf{P}^1$ be the ruling whose fibres are the residual lines, and put $Z = \{s_1, s_2\}_{\mathrm{red}}$. We enlarge T bad-reduction places of Z and the exceptions for this ruling and its boundary fibres. At each newly added place v, set $W_v = \mathcal{C}_v(\mathscr{Q}, \Delta_m)$. The set $f(W_v)$ is open since f is smooth. Propositions 2.5 give a strict T-Campana point t for $(\mathbf{P}^1, (1 - 1/m)Z)$ such that $t \in f(W_v)$ for any $t \in T$. By weak approximation on C_t, we may find a rational point R in every W_v. The point $P = \ell \cap C_t$ has infinite multiplicity with ℓ; outside T, its multiplicities with the other boundary components are those of t with Z. Thus P satisfies the required local conditions at $v \notin T$. The tuple equal to R at T and to P elsewhere belongs to $(C, (1 - 1/m)F)(\mathbf{A}_k)$ and maps into $\mathcal{N}$, the proof then follows from Lemma 3.1. □

For the rest of the paper, we may assume $E(k) = \varnothing$. Then $q|_W$ is nonzero and nondegenerate since the zero form would give $E = L$ with rational points, and a nonzero radical would give the rational vertex $\mathbf{P}(\mathrm{rad}(q|_W)) \subset E$. Thus E is smooth. If it is finite, it is a nonsplit degree-two point. If it has positive dimension, it is a smooth geometrically integral quadric.

4. Quadric surfaces

Let $Q \subset \mathbf{P}^3_k$ be a smooth quadric surface, and let Π_1, Π_2 be arbitrary hyperplanes defined over k. Put

$$H_i = Q \cap \Pi_i, \qquad E = H_1 \cap H_2, \qquad D = H_1 + H_2, \qquad \Delta_m = (1 - 1/m)D_{\mathrm{red}}.$$

The sections may coincide or share a line. A singular section is the union of two distinct geometric lines. Each distinct prime component has weight m, with the local Campana conditions imposed componentwise.

Proposition 4.1. *For any two hyperplane sections H_1, H_2 and every integer $m \geq 2$, (Q, Δ_m) satisfies Campana strong approximation.*

It suffices to prove the proposition when E is finite. Indeed, if $\Pi_1 \neq \Pi_2$ and E has positive dimension, then $E = \Pi_1 \cap \Pi_2$ is a k-line contained in Q, so Proposition 3.2 applies. If $\Pi_1 = \Pi_2$, choose an auxiliary hyperplane section F with $H_1 \cap F$ finite. Once the finite-intersection case is proved, applying it to $H_1 + F$ and then monotonicity [Wei26a, Lemma 2.5(1)] gives the result for $D_{\mathrm{red}} = H_1$.

The finite case with $E(k) \neq \varnothing$ also follows from Proposition 3.2. For the remaining argument, assume that E is a nonsplit closed point of degree two. Then D is reduced, so $\Delta_m = (1 - 1/m)D$.

4.1. **The conic pencil.** Let $\Lambda = \Pi_1 \cap \Pi_2$. The planes through Λ give a pencil of conics on Q with base locus E. Let $\pi : X = \mathrm{Bl}_E(Q) \to Q$. The pencil resolves to a morphism $f : X \to \mathbf{P}^1_k$. Its boundary has two vertical fibers, the strict transforms of H_1 and H_2, and a horizontal exceptional divisor of degree two over the base. Every smooth member C satisfies

$$H_1|_C = H_2|_C = E. \tag{4.1}$$

More precisely, write $E_{\overline{k}} = \{A, B\}$. If H_i is singular, its two geometric lines $L_{i,A}$ and $L_{i,B}$ satisfy

$$L_{i,A}|_C = A, \qquad L_{i,B}|_C = B.$$

A smooth section restricts to $A + B$. Thus the inclusion $(C, (1 - 1/m)E) \hookrightarrow (Q, \Delta_m)$ is an orbifold morphism, and Proposition 2.4 gives a continuous map on Campana spaces.

Write $K = k(E)$, a quadratic extension of k. The nondegenerate binary quadratic form on Λ is a scalar multiple of the norm form of K/k. We have the normal form

$$Q: \quad N_{K/k}(u) = p(t, w) = at^2 + btw + cw^2, \qquad H_1 = (t = 0), \quad H_2 = (w = 0), \tag{4.2}$$

where $u \in K$ represents a pair of k-coordinates. Smoothness of Q gives

$$b^2 - 4ac \neq 0. \tag{4.3}$$

The sections H_1 and H_2 are singular exactly when $c = 0$ and $a = 0$, respectively. The member of parameter $\tau = t/w$ is the conic

$$C_\tau: \quad N_{K/k}(u) = p(\tau, 1)w^2, \qquad t = \tau w. \tag{4.4}$$

Let $Z \subset \mathbf{P}^1$ be the reduced degree-two scheme of $p(t, w) = 0$. It indexes the two singular geometric fibres. Over a closed point $s \in Z$, the singular fiber splits by $K.k(s)$. Define the reduced marked set

$$M = \{0, \infty\} \cup Z, \qquad r = \deg M \in \{2, 3, 4\}.$$

Here $r = 4, 3, 2$ according as neither, exactly one, or both boundary sections are singular.

4.2. **Twisted norm auxiliary varieties.** Assume that E is nonsplit, with splitting field K, and use the reduced marked set M defined above. Write

$$Z \setminus \{0, \infty\} = \coprod_j \operatorname{Spec} F_j. \tag{4.5}$$

For each F_j, choose a quadratic field A_j/F_j containing K: take F_jK when K does not embed in F_j, and any quadratic extension otherwise. Assign to $s \in M$ the extensions $L_s/k(s)$ given by K/k at the boundary parameters and by A_j/F_j at the remaining singular parameters. If $r \leq 3$, all marked points are rational and $L_s = K$ for every $s \in M$. Each L_s splits the multiplicity-one component of $f^{-1}(s)$ since it contains K, as permitted in [HWW22, Theorem 4.1].

Choose an auxiliary affine coordinate in which M stays away from infinity, and write $a_s \in k(s)$ for the coordinate of s. For arbitrary $b_s \in k(s)^\times$, let $p_b : W_b \to \mathbf{P}^1$ be the standard auxiliary variety of [HWW22, Section 3.1], with equations

$$N_{L_s/k(s)}(z_s) = b_s(\lambda - a_s\mu), \qquad s \in M, \qquad p_b = [\lambda : \mu]. \tag{4.6}$$

Delete $\lambda = \mu = 0$ and the norm-singular loci specified in that construction. The base map is understood in the original pencil coordinates after undoing the auxiliary change of coordinates.

Lemma 4.2. *For every choice of the nonzero coefficients b_s, the variety W_b satisfies strong approximation off any finite place of k.*

Proof. Over $\overline{k}$, the norm equations have r blocks:

$$z_{j,+}z_{j,-} = \gamma_j(\lambda - a_j\mu), \qquad 1 \leq j \leq r, \tag{4.7}$$

with $\gamma_j \neq 0$ and distinct a_j.

Suppose first that $r = 4$. Eliminating λ, μ gives an affine cone over an intersection $V \subset \mathbf{P}^7$ of two quadrics. Its pencil has generic rank eight and exactly four rank-six members. The quadrics have no common factor, by[CTSSD87a, Lemmas 1.11 and 1.12], V is geometric integral and not a cone. The Jacobian criterion gives eight singular points. The standard deleted norm-singular loci have codimension at least two, and W_b is the remaining open subset of the smooth cone. Strong approximation then follows from Theorem 2.6.

If $r = 3$, relabel the singular boundary section as H_1. Then $p(t, w) = a\,t(t - \beta w)$ with $a\beta \neq 0$, and the marked parameters are $0, \infty, \beta$. In these coordinates the auxiliary equations are

$$N_{K/k}(x) = b_0\lambda, \qquad N_{K/k}(y) = b_\infty\mu, \qquad N_{K/k}(z) = b_\beta(\lambda - \beta\mu).$$

Eliminating λ, μ gives the nondegenerate rank-six quadric cone

$$N_{K/k}(z) = \frac{b_\beta}{b_0}N_{K/k}(x) - \frac{\beta b_\beta}{b_\infty}N_{K/k}(y).$$

Its projectivization is a smooth quadric of dimension four. The standard deleted loci have codimension at least two, so [HWW22, Lemma 6.4] gives strong approximation.

If $r = 2$, then $p(t, w) = b\,tw$ with $b \neq 0$. The equations are $N_{K/k}(x) = b_0\lambda$ and $N_{K/k}(y) = b_\infty\mu$. Eliminating λ, μ identifies W_b with an open subset of $\mathbf{A}^4_k$. The standard

deleted loci have codimension at least two, and strong approximation follows from [Wei21, Lemma 2.1]. □

4.3. **Choosing a fiber and treating new places.**

Lemma 4.3 (Choice of a fiber)**.** *Assume that E is nonsplit. For every nonempty open subset $\mathcal{N} \subset (Q, \Delta_m)(\mathbf{A}_k)$, there is $\tau \in (\mathbf{P}^1 \setminus M)(k)$ such that, for $j_{\tau,\mathbf{A}} : C_\tau \hookrightarrow Q$ and $\delta_\tau = (1 - 1/m)E$, the set*

$$j_\tau^{-1}(\mathcal{N}) \cap (C_\tau, \delta_\tau)(\mathbf{A}_k)$$

is nonempty.

Proof. Choose $(x_v) \in \mathcal{N}$. The Hasse principle gives $Q(k) \neq \varnothing$, so Q is k-rational and $\operatorname{Br} Q = \operatorname{Br} k$. Apply the descent argument in [HWW22, Lemma 4.2] to $f : X = \mathrm{Bl}_E(Q) \to \mathbf{P}^1_k$, with $A = 0$ and $K_s = L_s$, using Lemma 2.7 for its formal-lemma step. Removing M, the auxiliary infinity point from the base and the exceptional divisor on X, then we obtain an open subset $Q^\circ \subset Q$. The finite group of cup products used in the descent argument meets $\operatorname{Br} Q$ only in constants, so orthogonality is automatic. We obtain $(x'_v) \in \mathcal{N} \cap Q^\circ(\mathbf{A}_k)$, coefficients b_s, and lifts $(z_v) \in W_b(\mathbf{A}_k)$ with $p_b(z_v) = f(x'_v)$.

Shrink $\mathcal{N}$ about (x'_v) to a basic open set. Choose a finite set of places T containing the local restrictions of $\mathcal{N}$, all geometric and reduction exceptions, the places ramified in K/k, the places of bad reduction of the norm model, the nonunit places of the defining and twisting coefficients, and the places where the lifts are not integral. At T, choose neighborhoods of x'_v small enough to retain membership in $\mathcal{N}$. Choose a finite place $v_0 \notin T$ splitting completely in the normal closures of all L_s. Lemma 4.2 supplies a rational point on W_b, arbitrarily close to z_v at T and integral outside $T \cup \{v_0\}$, whose image τ in $\mathbf{P}^1_k$ avoids M. Smoothness gives points on C_τ in the chosen neighborhoods at T.

If $v \notin T \cup \{v_0\}$ and τ reduces to a marked point s, the induced place of $k(s)$ has degree one over v, since the marked points have étale reduction and τ has rational reduction. By [HWW22, Proposition 3.8], it extends to a place of L_s of degree one over v. Since K embeds in L_s, the place v splits in K. Splitting at v_0 is automatic. At a split place outside T, both points of E are k_v-rational. We choose a local point sufficiently close to either of $E(k_v)$. This point may have arbitrarily large intersection multiplicities with the boundary components passing through that point, and zero intersection multiplicity with the remaining components. It therefore satisfies the Campana conditions. At inert places outside T, the reduction of τ avoids M, so τ and $p(\tau, 1)$ are units. By surjectivity of the unramified norm on units, we may choose an integral solution u of (4.4) with $w = 1$. In the point $[u : \tau : 1]$, both boundary coordinates are units, so both boundary multiplicities are zero. These choices map into $\mathcal{N}$ and are Campana on (C_τ, δ_τ) at almost every place, by (4.1). They therefore give a point of the required inverse image. □

Proof of Proposition 4.1. By the reductions above, we may assume that E is a nonsplit closed point of degree two. Let $\mathcal{N} \subset (Q, \Delta_m)(\mathbf{A}_k)$ be nonempty and open. Lemma 4.3 supplies a smooth conic whose Campana adelic inverse image of $\mathcal{N}$ is nonempty. Lemma 3.1 gives a strict rational point in $\mathcal{N}$, the density then follows. □

Corollary 4.4. *Let Σ be a smooth quadric surface and B a hyperplane section. Then $(\Sigma, (1-1/m)B)$ satisfies Campana strong approximation for every $m \geq 2$.*

Proof. Apply Proposition 4.1 with $H_1 = H_2 = B$. □

5. Reduction to surfaces

The case $d = 2$ was proved in Proposition 4.1, and the case $E(k) \neq \varnothing$ follows from Proposition 3.2. Throughout this section, we assume $d \geq 3$ and $E(k) = \varnothing$. We reduce the finite and positive-dimensional cases to smooth quadric surfaces with two and one hyperplane boundary sections, respectively.

5.1. **Finite common intersection.** First assume that E is finite, hence reduced. Let

$$\Lambda_0 = \Pi_2 \cap \cdots \cap \Pi_d \simeq \mathbf{P}^2_k, \qquad B = Q \cap \Lambda_0, \qquad E = B \cap \Pi_1. \tag{5.1}$$

The restriction of the defining quadratic form to the line $\Lambda = \Lambda_0 \cap \Pi_1$ has rank two. Hence B is a reduced plane conic, either smooth or a pair of geometric lines. In the latter case its vertex O is k-rational and lies outside H_1, since E is reduced.

For the linear algebra below, we write $Q \subset \mathbf{P}(V)$, where V is a k-vector space of dimension $d+2$ and q is its nondegenerate quadratic form. Choose linear forms ℓ_i defining Π_i, and put

$$V_i = \ker(\ell_i), \qquad V_0 = \bigcap_{i=2}^{d} V_i, \qquad W_0 = V_0 \cap V_1.$$

Then $\Lambda_0 = \mathbf{P}(V_0)$, $\Pi_1 = \mathbf{P}(V_1)$ and $\Lambda = \mathbf{P}(W_0)$, with $\dim V_0 = 3$ and $\dim W_0 = 2$. Reducedness of E makes $q|_{W_0}$ nondegenerate.

Lemma 5.1. (1) *There is a nonempty Zariski-open subset $\mathcal{U} \subset Q \setminus D$ such that, for every $R \in \mathcal{U}(k)$,*

$$\Pi_R = \langle \Lambda_0, R \rangle \simeq \mathbf{P}^3_k, \qquad \Sigma_R = Q \cap \Pi_R \tag{5.2}$$

is a smooth quadric surface.

(2) *For every R chosen in (1), the first boundary section $F_1 = H_1|_{\Sigma_R}$ is a reduced plane conic, and the boundary restrictions satisfy*

$$H_i|_{\Sigma_R} = B \quad (2 \leq i \leq d), \qquad F_1 \cap B = E \tag{5.3}$$

scheme-theoretically. In particular, F_1 and B have no common component.

Proof. (1) The parameter space $\mathcal{P} = \mathbf{P}(V/V_0)$ parametrizes four-dimensional subspaces $U \supset V_0$. The form $q|_{V_0}$ has rank three or two. In the first case, choose $z \in V_0^{\perp}$ with $q(z) \neq 0$; in the second, let o span its radical and choose z with $b_q(o, z) \neq 0$, where b_q is the polar form. In either case $q|_{V_0 + kz}$ has rank four. Thus smooth surface sections form a nonempty open subset of $\mathcal{P}$.

Projection from Λ_0 gives

$$Q \setminus \Lambda_0 \longrightarrow \mathcal{P}, \qquad [r] \longmapsto V_0 + kr.$$

It is dominant, since every general smooth surface section contains geometric points outside B. The inverse image of the smooth-section locus under this projection, intersected with $Q \setminus D$, is the required nonempty open subset $\mathcal{U}$.

(2) For $R = [r] \in \mathcal{U}(k)$, put $U_R = V_0 + kr$. For $i \geq 2$, ℓ_i vanishes on V_0 and $\ell_i(r) \neq 0$, because $R \notin D$. Hence $V_i \cap U_R = V_0$, giving $H_i|_{\Sigma_R} = Q \cap \mathbf{P}(V_0) = B$ and $F_1 \cap B = B \cap \Pi_1 = E$ scheme-theoretically. A hyperplane restriction of the rank-four form defining the smooth surface Σ_R has rank at least two, so F_1 is reduced. Finally, finiteness of E excludes a common component of F_1 and B. □

Remark 5.2. The section F_1 may be singular. Its smoothness defines an open condition on $R \in Q \setminus D$. This locus is nonempty by the same nondegeneracy and projection arguments as in the proof of Lemma 5.1, applied to plane sections of $Q \cap \Pi_1$ through Λ. Indeed, $q|_{W_0}$ is nondegenerate and $\operatorname{rank}(q|_{V_1}) \geq d \geq 3$. Denote this nonempty open subset by $\mathcal{U}_1$. Since Q is irreducible, $\mathcal{U} \cap \mathcal{U}_1$ is nonempty and open. Thus the smoothness of F_1 may also be imposed if desired. Reducedness of $E = F_1 \cap B$ forces the vertex of either singular conic to lie outside the other. Both conics are therefore smooth at E and meet transversely there. The extra smoothness of F_1 and the separate transversality assertion are not required for Proposition 4.1 to apply.

The following lemma gives the nonempty adelic set needed on the chosen surface.

Lemma 5.3. *Let $\mathcal{N} \subset (Q, \Delta_m)(\mathbf{A}_k)$ be nonempty and open. There exists $R \in \mathcal{U}(k)$ such that, with $\Sigma = \Sigma_R$, $\delta_R = (1 - 1/m)(F_1 + B)$ and $j : \Sigma \hookrightarrow Q$, the set*

$$\mathcal{G} = j_{\mathbf{A}}^{-1}(\mathcal{N}) \cap (\Sigma, \delta_R)(\mathbf{A}_k) \tag{5.4}$$

is nonempty.

Proof. Shrink $\mathcal{N}$ to a basic open subset, taking its finite set T of local conditions large enough to contain the model and reduction exceptions for B and E. The Hasse principle and weak approximation give $R \in \mathcal{U}(k)$ satisfying the local conditions at T.

If $v \notin T$, we lift a residue-field point outside E when B is smooth, and otherwise take its vertex. Hence we may choose $b_v \in B(k_v)$ reducing outside H_1. Since these points lie on H_i for $i \geq 2$ and have zero multiplicity with H_1, they satisfy the required Campana conditions on Q. The tuple equal to R at T and to b_v elsewhere belongs to $(\Sigma, \delta_R)(\mathbf{A}_k)$, since it lies on B and has zero multiplicity with F_1 at almost every place. Its image lies in $\mathcal{N}$, proving the assertion. □

Remark 5.4. Finiteness of E implies that the hyperplanes are linearly independent and that $\deg E = 2$. Our assumption $E(k) = \varnothing$ makes E reduced, which forces the intermediate plane conic B to be reduced, although it may be singular. A nonreduced E would have the form $2P$ with $P \in Q(k)$, a case already covered by Proposition 3.2, for this case B may even be a double line. Thus no separate reducedness or smoothness hypothesis is needed in the finite-intersection argument.

5.2. **Positive-dimensional common intersection.** Now assume that $\dim E \geq 1$. With $L = \mathbf{P}(W)$ as in Section 3, the form $q|_W$ is anisotropic over k. Choose a k-plane

$L' = \mathbf{P}(V') \subset L$ and put $B = Q \cap L'$. The restriction $q|_{V'}$ is nondegenerate, since a nonzero radical vector would be isotropic. Thus B is a smooth conic.

The rank-three argument in the proof of Lemma 5.1(1) gives a nonempty open subset $\mathcal{U}_+ \subset Q \setminus D$ such that

$$\Sigma_R = Q \cap \langle L', R \rangle \tag{5.5}$$

is a smooth quadric surface for every $R \in \mathcal{U}_+(k)$. Since $R \notin D$ and $L' \subset \Pi_i$, we have

$$H_i|_{\Sigma_R} = B \qquad (1 \le i \le d). \tag{5.6}$$

Thus the reduced boundary on Σ_R is the smooth conic B.

Proof of Theorem 1.1. Let $\mathcal{N} \subset (Q, \Delta_m)(\mathbf{A}_k)$ be nonempty and open.

For finite E, Lemma 5.3 gives a smooth surface Σ with boundary $\delta = (1 - 1/m)(F_1 + B)$ and nonempty $\mathcal{G} = j_{\mathbf{A}}^{-1}(\mathcal{N}) \cap (\Sigma, \delta)(\mathbf{A}_k)$. For positive-dimensional E, choose a basic subset of $\mathcal{N}$ with finite set T of local conditions, large enough that $B(k_v) \neq \varnothing$ outside T. The Hasse principle and weak approximation choose $R \in \mathcal{U}_+(k)$ satisfying these local conditions. Put $\Sigma = \Sigma_R$ and $\delta = (1 - 1/m)B$. The tuple equal to R at T and to points of $B(k_v)$ elsewhere belongs to $(\Sigma, \delta)(\mathbf{A}_k)$ and maps into $\mathcal{N}$. Thus $\mathcal{G}$ is again nonempty.

In both cases $j : (\Sigma, \delta) \hookrightarrow (Q, \Delta_m)$ is an orbifold morphism, so $\mathcal{G}$ is open by Proposition 2.4. Proposition 4.1, respectively Corollary 4.4, gives a strict rational point in $\mathcal{G}$. Its image is strict on Q and belongs to $\mathcal{N}$, proving density. □

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